

Deep Learning Model Performance Comparison
Neural Network Architecture Benchmark

Model Name	Parameters (M)	Training Time (h)	Inference Time (ms)	ImageNet Accuracy (%)	ImageNet mAP (%)	COCO mAP (%)	Memory (GB)
ResNet-50	25.5	48.2	1250	76.1	92.9	37.2	2.1
ResNet-101	44.5	72.5	890	77.8	93.7	39.1	3.2
EfficientNet-B0	5.3	36.1	2150	77.1	93.5	38.4	0.9
EfficientNet-B3	19.3	98.5	1050	82.9	96.3	42.8	2.8
Vision Transformer-B/16	86.6	156	850	81.1	95.3	40.5	4.2
YOLO-v8n	3.2	24	3500	72.5	90.1	35.6	0.6
YOLO-v8s	11.2	42	2800	75.2	91.8	37.9	1.2
YOLO-v8m	25.9	68	1950	79.3	94.1	40.2	2.3
Swin-Transformer-TL	28.3	120	720	81.5	95.5	41.1	3.5
ConvNeXt-B	28.6	110	950	82.1	95.9	41.8	3.3

Model Performance Analysis

Model	Params (M)	Accuracy (%)	Efficiency (FLOPs)	Params-per-Speed Ratio	Accuracy (Memory per Accuracy)
ResNet-50	25.5	76.1	2.984314	0.977011	0 1.465517
ResNet-10	44.5	77.8	1.748315	1.600719	1.7 2.401079
EfficientNe	5.3	77.1	14.54717	0.195572	1 0.293358
EfficientNe	19.3	82.9	4.295337	0.586626	6.8 0.879939
Vision Tra	86.6	81.1	0.93649	2.784566	5 4.176849
YOLO-v8n	3.2	72.5	22.65625	0.142222	-3.6 0.213333
YOLO-v8s	11.2	75.2	6.714286	0.444444	-0.9 0.666667
YOLO-v8m	25.9	79.3	3.061776	0.883959	3.2 1.325939
Swin-Trans	28.3	81.5	2.879859	0.898413	5.4 1.347619
ConvNeXt	28.6	82.1	2.870629	0.890966	6 1.336449

Training Progress - Loss and Accuracy

Epoch	Training L	Validation	Training A	Validation	Learning R	Overfitting Gap (Val-Train)
1	2.302	2.298	12.5	15.2	0.001	-0.004
2	1.956	1.845	28.3	35.1	0.001	-0.111
3	1.542	1.423	42.7	48.9	0.001	-0.119
5	0.892	0.934	65.3	62.1	0.001	0.042
10	0.345	0.412	87.2	84.5	0.0005	0.067
20	0.156	0.198	94.2	91.3	0.0001	0.042
30	0.089	0.127	96.5	93.8	0.00005	0.038
40	0.052	0.098	97.8	94.9	0.00001	0.046
50	0.031	0.084	98.6	95.5	0.000001	0.053

Hyperparameter Tuning Results

Experiment	Learning Rate	Batch Size	Weight Decay	Warmup Steps	Final Accuracy	Converger	Accuracy per Hour
1	0.001	32	0.0001	500	91.2	48	1.9
2	0.001	64	0.0001	500	92.1	45	2.046667
3	0.001	128	0.0001	500	91.8	42	2.185714
4	0.002	64	0.0001	1000	93.5	46	2.032609
5	0.002	64	0.00005	1000	93.2	47	1.982979
6	0.0005	64	0.0001	500	92.8	52	1.784615
7	0.002	128	0.0001	1000	94.1	44	2.138636
8	0.002	128	0.00005	1500	94.8	48.5	1.954639